Hypothesis Testing II

Unit - 5

Session - 9

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Agenda

- Test of Correlation
- Chi-square test
  - Goodness of Fit
  - Independence of Attributes
- One Way - ANOVA

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Test of Correlation

A researcher wants to investigate if there is a linear relationship between the number of hours studied and the scores obtained by students in an exam. A sample of 10 students provides the following data. Test if there is a significant positive correlation between hours studied and exam scores using the Pearson correlation coefficient at a 5% significance level ($\alpha=0.05$)

Hours Studied (X)	Exam Scores (Y)
2	50
3	55
4	60
5	65
6	70
7	75
8	80
9	85
10	90
11	95

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Formulate Hypotheses:

- Null Hypothesis (H0): There is no correlation between hours studied and exam scores, i.e., $\rho=0$
- Alternative Hypothesis (Ha): There is a significant positive correlation between hours studied and exam scores, i.e., $\rho \neq 0$

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Correlation Formula :

$$corr_{xy} = \frac{\overbrace{\boxed{\sum_{i=1}^{n}(x_i - \overline{x})\,(y_i - \overline{y})}}^{\text{Codeviance}}}{\underbrace{\sqrt{\boxed{\sum_{i=1}^{n}(x_i - \overline{x})^2}}}_{\text{Sum of Squared Deviations (x)}}\ \underbrace{\sqrt{\boxed{\sum_{i=1}^{n}(y_i - \overline{y})^2}}}_{\text{Sum of Squared Deviations (y)}}}$$

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Pearson Correlation Formula :

The formula for the **Pearson correlation coefficient** r is given by:

$$r = \frac{n\sum xy - \sum x \sum y}{\sqrt{[n\sum x^2 - (\sum x)^2]\,[n\sum y^2 - (\sum y)^2]}}$$

Where:

- n is the number of data points (sample size).
- x and y are the variables being compared.
- $\sum xy$ is the sum of the product of corresponding values of x and y.
- $\sum x$ is the sum of all values of x.
- $\sum y$ is the sum of all values of y.
- $\sum x^2$ is the sum of the squares of all values of x.
- $\sum y^2$ is the sum of the squares of all values of y.

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Pearson Correlation Test:

Formulate Hypotheses:

- Null Hypothesis (H0): There is no correlation between hours studied and exam scores, i.e., ρ=0
- Alternative Hypothesis (Ha): There is a significant positive correlation between hours studied and exam scores, i.e., ρ > 0

Calculation :

E x = 65

E y = 680

E x^2 = 505

E y^2 = 48850

E xy = 4805

n=10

r = (10 * 4805) - (65 * 680) / sqrt(((10 * 505) - (65^2)) * ((10 * 48850)- (680^2))

r = 0.8296

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3. Critical Value:

- Degrees of freedom df=10−2=8.
- For a one-tailed test with α=0.05, the critical value from Pearson's table for 8 degrees of freedom is approximately critical=0.632 (for ρ≠0).

4. Compare the Test Statistic:

- The calculated r=0.826 which is **less than the critical value** 0.632.
- In fact, it's negative, indicating no positive correlation.
- Thus, we **reject H0**

5. Conclusion :

There is **significant positive correlation** between the number of hours studied and the exam scores based on the sample data. The result is not statistically significant at the 5% significance level.

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Table

A researcher wants to investigate if there is a linear relationship between the age and blood pressure . A sample of 6 patients provides the following data. Test if there is a significant positive correlation between age and blood pressure using the Pearson correlation coefficient at a 5% significance level (α=0.05)

Age (x)	Blood Pressure (Y)
25	120
30	125
35	130
40	135
45	140
50	150

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Pearson Correlation Test:

Formulate Hypotheses:

- Null Hypothesis (H0): There is no correlation between Age and blood pressure, i.e., ρ=0
- Alternative Hypothesis (Ha): There is a significant positive correlation between Age and blood pressure, i.e., ρ > 0

Calculation :

E x = 225

E y = 800

E x^2 = 8875

E y^2 = 107250

E xy = 30500

n=6

r = 6(30500)−(225)*(800) / sqrt((6(8875)−(225)^2) * ((6(107250)−800^2))

r = 0.9897433

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3. Critical Value:

- Degrees of freedom df=6−2=4.
- For a one-tailed test with α=0.05, the critical value from Pearson's table for 8 degrees of freedom is approximately critical=0.811 (for ρ≠0).

4. Compare the Test Statistic:

- The calculated r=0.99 which is **greater than the critical value** 0.811.
- Thus, we **reject H0**

5. Conclusion :

There is **significant positive correlation** between the age and blood pressure based on the sample data. The result is not statistically significant at the 5% significance level.

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Tests for Categorical Data

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Tests for categorical data

- The collected data is at times best represented by categories
- These categories are summarized by their frequency of occurrence. It may be of interest whether this frequency is equal to the expectation/claim
- It may also be of interest to know whether the categories are statistically independent

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Chi-square tests

- These interests are tested by non-parametric ways
- Tests based on the chi-square distribution are used
- The chi-square tests are used to test:
  - The goodness of fit
  - The independence of two attributes
- Chi-square tests are also used to test for population variance

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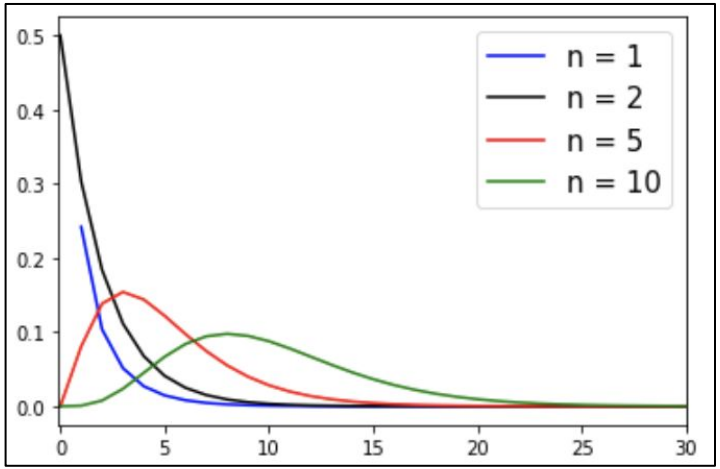
Chi-square distribution

- Chi-square distribution is sum of squared of standard normals

  Let $X_1, X_2, \ldots, X_n$ be n standard normal variates,

  then $Y = X^2_1 + X^2_2 + \ldots + X^2_n$,

  Y follows χ^2 distribution with *n* degrees of freedom.



- The mean of the distribution is *n* and its variance is *2n*

- The distribution is positively skewed

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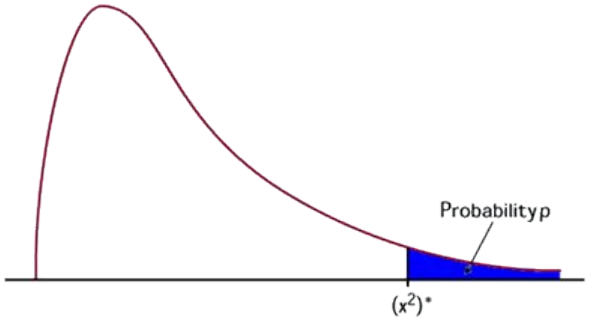
Chi Square (χ^2) Test

- The Chi Square test is a non-parametric Test
- It tests relationship between two nominal variables
- The statistic chi square is given by the equation and has a distribution as shown.

$$\sum \frac{(O_i - E_i)^2}{E_i}$$



The conditions for using chi square

- Data must be counts
- The observations must be random and independent
- None of the cells must have a count less than 5

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Properties of Chi-Square Distribution

1. The mean and standard deviation of a chi-square distribution are k and $\sqrt{2k}$ respectively, where k is the degrees of freedom.

2. As the degrees of freedom increases, the probability density function of a chi-square distribution approaches normal distribution.

3. Chi-square goodness of fit is one of the popular tests for checking whether a data follows a specific probability distribution.

4. Chi square test is a right tailed test.

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χ^2 Test for Goodness of Fit

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Chi-square Goodness of fit tests

- Goodness of fit tests are hypothesis tests that are used for comparing the observed distribution of data with expected distribution of the data to decide whether there is any statistically significant difference between the observed distribution and a theoretical distribution (for example, normal, exponential, etc.) based on the comparison of observed frequencies in the data and the expected

Null hypothesis:	***There is no statistically significant difference between the observed frequencies and the expected frequencies from a hypothesized distribution***
Alternative hypothesis	***There is statistically significant difference between the observed frequencies and the expected frequencies from a hypothesized distribution.***

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Chi-square Goodness of fit tests

$$\chi^2 = \sum_{i=1}^{n} \sum_{j=1}^{m} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

where O_{ij} is the observed frequency in category (i, j)

E_{ij} is the expected frequency in category (i, j)

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Chi-Square One Way - Example

- **A1 airlines operated daily flights to several Indian cities. The operations manager believes that 30% of their passengers prefer vegan food, 45% prefer vegetarian food , 20% prefer non-veg food 5% request for Jain food.**

- **A sample of 500 passengers was chosen to analyse the food preferences**

Food type	Vegan	Vegetarian	Non-Vegetarian	Jain
Number of passengers (obs)	190	185	90	35

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Solution

Null hypothesis : Probability distribution of the food preference is

P(Vegan) = 0.30, P(Veg) = 0.45

P(Non-veg) = 0.20 P(Jain) = 0.05

Alternative hypothesis : Probability distribution of the food preference is not defined in the null hypothesis.

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Chi-Square - Calculation

Foof Type	Actual count (Observed)	Allocation by Ops manager (Expected)	Difference	(obs -exp)^2	(obs -exp)^2 / Expected
Vegan	190	150	40	1600	10.67
Vegetarian	185	225	-40	1600	7.11
Non-Vegetarian	90	100	-10	100	1
Jain	35	25	10	100	4
				Chi observed	**22.78**

df = 4 -1 = 3

$$\chi^2_{(3,0.05)} = 7.814728$$

Since the chi-square calculated value > critical value, we reject the null hypothesis.

We conclude that probability distribution of the food preference is not defined in the null hypothesis

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Chi-square test for goodness of fit

At an emporium, the manager is interested in knowing age group which visits the mall during the day. He defines categories as - children (age < 13), teenagers ($13 \leq$ age < 20), adults ($20 \leq$ age < 55) and senior citizens ($55 \leq$ age). Moreover, he wishes to plan his inventory of goods accordingly.

He claims that out of all the people who visited 5% are children, 38% are teenagers, 2% are senior citizens are remaining are adults.

Can the owner verify the managers claim?

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Chi-square test for goodness of fit

- The hypothesis to test whether the data fits the a specified distribution

H_0: There is no difference between observed frequencies and expected frequencies

Against

H_1: There is difference between observed frequencies and expected frequencies

- Failing to reject H_0, implies that there is no difference between observed frequencies and expected frequencies

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Chi-square test for goodness of fit

- The test statistic is given by

$$\chi^2 = \sum_{i=1}^{k} \frac{O_i^2}{e_i} - N$$

observed frequency

total number of observations

estimated frequency

- Under H_0, the test statistics follows χ^2 distribution with k-p-1 d.f

  where *k*: number of class frequencies

  p: number of parameter estimated for fitting

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Ref: Test statistic for goodness of fit (A.1)

Chi-square test for goodness of fit

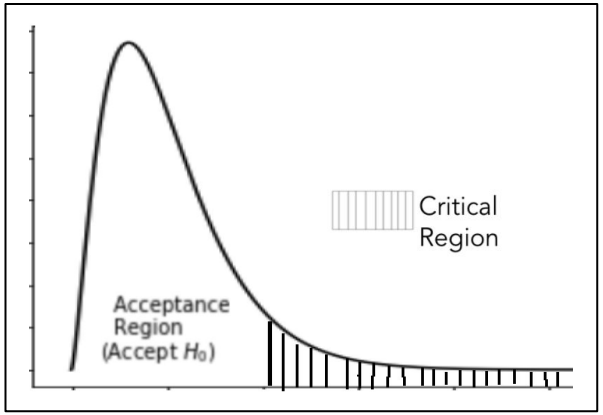
Decision Rule:

Reject H_0 if $\chi^2_{calc} \geq \chi^2_{k-p-1,\alpha}$

or

Reject H_0 if p-value $\leq \alpha$

Where, α is the level of significance (l.o.s.)



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Test for goodness of fit

Question:

At an emporium, the manager is interested in knowing age group which visits the mall during the day. He defines categories as - children, teenagers, adults and senior citizens. He plans to have his inventory of goods accordingly. He claims that out of all the people who visited 5% are children, 38% are teenagers, 2% are senior citizens are remaining are adults.

From a sample of 180 people it was seen that 25 were children, 50 were teenagers, 90 were adults and 15 were senior citizens

Test the manager's claim at 95% confidence level.

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Test for goodness of fit

Solution:

We can tabulate the given data as follows:

	Manager Claimed Frequency	The frequency expected from 180 customers (e_i)	The frequency observed from 180 customers (O_i)
Children	5%	0.05 x 180 = 9	25
Teenagers	38%	0.38 x 180 = 68.4 ≅ 68	50
Adults	55%	0.55 x 180 = 99	90
Senior Citizens	2%	0.02 x 180 = 3.6 ≅ 4	15

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Test for goodness of fit

Solution:

To test, H_0: The managers claim is correct Against H_1: The managers claim is false

The test statistic is $\chi^2 = \sum_{i=1}^{k} \frac{O_i^2}{e_i} - N$

$$= [\frac{25^2}{9} + \frac{50^2}{68} + \frac{90^2}{99} + \frac{15^2}{4}] - 180$$

$$= 64.27$$

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Test for goodness of fit

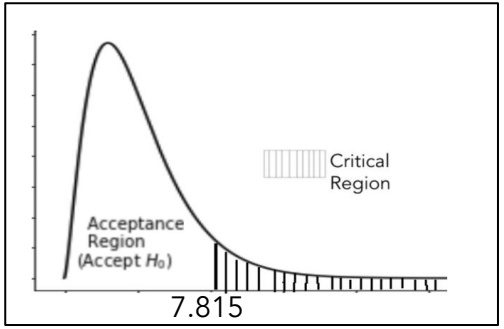
Solution:

Here there are 4 class frequencies, i.e k = 4. Since no parameter was calculated p = 0

From the statistical table for χ^2 distribution, $\chi^2_{k-p-1,\alpha} = \chi^2_{3,0.05} = 7.815$

The test statistic $\chi^2_{calc} = 64.27$

Since $\chi^2_{calc} > \chi^2_{k-p-1,\alpha}$, reject H_0.



The managers claim is false, his claim is different than what was observed from the data

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Table

- If the expected frequencies $e_i \geq 5$ and the total frequencies are large (≥ 50) the test can be used

- If $e_i < 5$, the class is merged with the neighbouring class for observed and expected frequencies until the it becomes ≥ 5

- It is not applicable for testing the goodness of fit of a straight line or any curve (exponential curve, second degree curve)

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χ^2 Test for Independence of Attributes

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Chi-square test for independence of attributes

- The hypothesis to test independence of attributes

  H_0: The attributes are independent against H_1: The attributes are dependent

- Failing to reject H_0, implies that the attributes are independent

- Decision rule: Reject H_0 at α l.o.s if $\chi^2_{(r-1)(s-1)} \geq \chi^2_{(r-1)(s-1);\alpha}$

  or

  Reject H_0 if p-value $\leq \alpha$

  Where, α is the level of significance (l.o.s.)

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Chi-square test for independence of attributes

- The test statistic is given by

$$\chi^2 = \sum_{i=1}^{r} \sum_{j=1}^{c} \frac{O_{ij}^2}{e_{ij}} - N$$

observed frequency in i^{th} row and j^{th} column

total number of observations

estimated frequency in i^{th} row and j^{th} column

- Under H_0, the test statistics follows χ^2 distribution with (r-1)(c-1) d.f where *r* levels for a category and *c* levels for another category

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Ref: Test statistic for independence of attributes (A.2)

Test for independence of attributes

Question:

A study was conducted to test the effect of the malaria parasite - plasmodium falciparum - on heterozygous and homozygous humans. The vaccine was given to a cohort of 252 humans. Test whether the heterozygous humans are better protected than homozygous.

	Infected with plasmodium falciparum	Not infected with plasmodium falciparum
Heterozygous humans	93	51
Homozygous humans	68	40

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Test for independence of attributes

Solution:

Let X: The zygote type - Homozygous or Heterozygous
Y: Whether infected or not with malaria parasite

Here X and Y are two attributes.

To test, H_0: The attributes are independent Against H_1: The attributes are dependent

Here there are 2 rows and 2 columns.

Let us computed the expected frequency.

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Test for independence of attributes

Solution:

In order to compute the expected frequency, first compute the total of the each column and row.

	Infected with plasmodium falciparum	Not infected with plasmodium falciparum	Total
Heterozygous humans	93	51	144
Homozygous humans	68	40	108
Total	161	91	252

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Test for independence of attributes

Solution:

The expected frequencies are computed as

$$e_{ij} = \frac{\text{Total}_{\text{row}} \times \text{Total}_{\text{column}}}{N}$$

$$e_{11} = \frac{144 \times 161}{252} = 92$$

	Infected with plasmodium falciparum	Not infected with plasmodium falciparum	Total
Heterozygous humans	$\frac{144 \times 161}{252} = 92$		144
Homozygous humans			108
Total	161	91	252

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Test for independence of attributes

Solution:

Similarly compute the expected frequencies for other classes

	Infected with plasmodium falciparum	Not infected with plasmodium falciparum	Total
Heterozygous humans	$\frac{144\times161}{252} = 92$	$\frac{144\times91}{252} = 52$	144
Homozygous humans	$\frac{108\times161}{252} = 69$	$\frac{108\times91}{252} = 39$	108
Total	161	91	252

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Test for independence of attributes

Solution:

The observed frequency (O_{ij})

	Infected with plasmodium falciparum	Not infected with plasmodium falciparum	Total
Heterozygous humans	93	51	144
Homozygous humans	68	40	108
Total	161	91	252

The expected frequency (e_{ij})

	Infected with plasmodium falciparum	Not infected with plasmodium falciparum	Total
Heterozygous humans	92	52	144
Homozygous humans	69	39	108
Total	161	91	252

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Test for independence of attributes

Solution:

The test statistic is computed as

$$\chi^2 = \sum_{i=1}^{r} \sum_{j=1}^{c} \frac{O_{ij}^2}{e_{ij}} - N$$

$$= \frac{93^2}{92} + \frac{51^2}{52} + \frac{68^2}{69} + \frac{40^2}{39} - 252$$

$$= 0.070$$

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Test for independence of attributes

Solution:

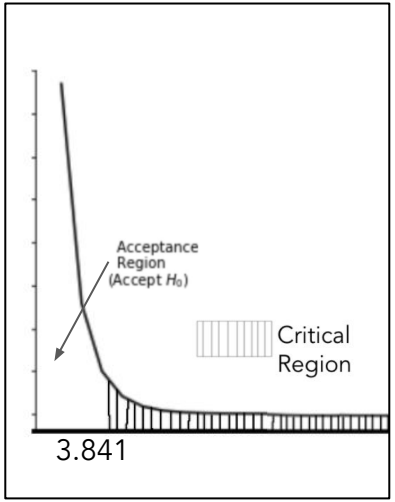
Here there are 2 levels of one attribute and 2 levels of another.
Thus the degrees of freedom are (r-1)(c-1) = (2-1)(2-1) = 1

From the statistical table for χ^2 distribution, $\chi^2_{(r-1)(s-1),\alpha} = \chi^2_{1,0.05} = 3.841$

The test statistic $\chi^2_{calc} = 0.070$

Since $\chi^2_{calc} < \chi^2_{k-p-1,\alpha}$, we fail to reject H_0.

The attributes are independent.



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Independence of attributes

Question:

A psychologist wants study whether the happiness quotient of children in the house is related to the family income. He collects data of 1300 children is there enough evidence to claim that they are related.

	Low income	Moderate income	High income
Happy	245	354	243
Unsatisfied	98	220	140

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Category	Low Income	Moderate Income	High Income
Happy (E_1)	(842×343)/1300 = 222.03	(842×574)/1300 = 371.56	(842×383)/1300 = 248.41
Unsatisfied (E_2)	(458×343)/1300 = 120.97	(458×574)/1300 = 202.44	(458×383)/1300 = 134.59

Calculate each term O_i^2/E_i:

- $245^2/222.03 \approx 270.3463$
- $354^2/371.56 \approx 337.2699$
- $243^2/248.41 \approx 237.7078$
- $98^2/120.97 \approx 79.3916$
- $220^2/202.44 \approx 239.0832$
- $140^2/134.59 \approx 145.6275$

Sum: $\sum \frac{O_i^2}{E_i} \approx 1309.4263$

$$\chi^2 = 1309.4263 - 1300 = \boxed{9.4263}$$

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- Degrees of freedom: $(r-1)(c-1) = (2-1)(3-1) = 2$.
- At $\alpha = 0.05$, $\chi^2_{crit}(df = 2) = 5.991$.

Since $\chi^2_{calc} \approx 9.43 > 5.991$, **reject** H_0.

There is **sufficient evidence** at the 5% level to conclude that **children's happiness and family income are related** (they are not independent).

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Tests based on Chi-squared distribution for categorical data are one tailed tests.

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ANALYSIS OF VARIANCE (ANOVA)

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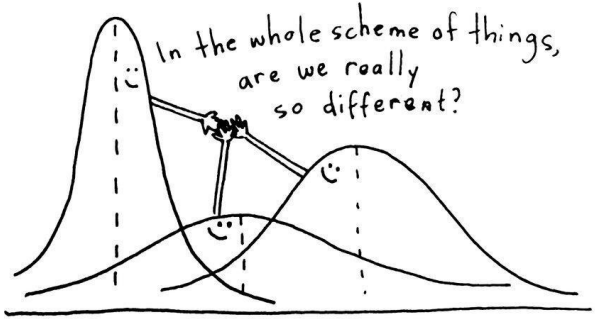
What is ANOVA and why is it used?



Source: Questionpro

Analysis of Variance (ANOVA) is a parametric statistical method used to compare datasets. This technique was invented by R.A. Fisher, and is thus often referred to as Fisher's ANOVA, He developed ANOVA while dealing with agronomic data as well.

ANOVA is best applied where more than two populations or samples are meant to be compared. The use of ANOVA involves certain **key assumptions**, including the following:

- **Independence of case**: Independence of case assumption means that the case of the dependent variable should be independent or the sample should be selected randomly. There should not be any pattern in the selection of the sample.

- **Normality**: Distribution of each group should be normal. The Kolmogorov-Smirnov or the Shapiro-Wilk test may be used to confirm normality of the group.

- **Homogeneity**: Homogeneity means variance between the groups should be the same. Levene's test is used to test the homogeneity between groups.

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T-Test and ANOVA : Difference

There is a thin line of demarcation amidst t-test and ANOVA, i.e. when the population means of only two groups is to be compared, the **t-test** is used, but when means of more than two groups are to be compared, **ANOVA** is preferred.

Comparison Chart

BASIS FOR COMPARISON	T-TEST	ANOVA
Meaning	T-test is a hypothesis test that is used to compare the means of two populations.	ANOVA is a statistical technique that is used to compare the means of more than two populations.
Test statistic	$(\bar{x}-\mu)/(s/\sqrt{n})$	Between Sample Variance/Within Sample Variance

★ t-test is a special type of ANOVA that can be used when we want to compare their means. Although the chances of errors might increase if t-test is used when we have to compare more than two means of the populations concurrently, that is why ANOVA is used.

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Terminologies related to ANOVA

Before staring with the applications of ANOVA, lets have a look at some of the common terminologies used in the technique:

- **Mean:** Mean is a simple or arithmetic average of a range of values. There are two kinds of means that we use in ANOVA calculation , $(\mu_1, \mu_2 \,\&\, \mu_3)$ which are separate sample means and the grand mean (μ). The grand mean is the mean of sample means or the mean of all observations combined, irrespective of the sample.

- **Hypothesis:** A hypothesis is an educated guess about something in the world around us. It should be testable either by experiment or observation. Just like any other kind of hypothesis that you might have studied in statistics, ANOVA also uses a Null hypothesis and an Alternate hypothesis. The Null hypothesis in ANOVA is valid when all the sample means are equal, or they don't have any significant difference. Thus, they can be considered as a part of a larger set of the population. On the other hand, the alternate hypothesis is valid when at least one of the sample means is different from the rest of the sample means. In mathematical form, they can be represented as:

$$H_o: \ \mu_1 = \mu_2 = \cdots = \mu_L \qquad \textit{Null hypothesis}$$

$$H_1: \ \mu_l \neq \mu_m \qquad \textit{Alternate hypothesis}$$

where μ_l and μ_m g to any two sample means out of all the samples considered for the test. In other words, the null hypothesis states that all the sample means are equal or the factor did not have any significant effect on the results. Whereas, the alternate hypothesis states that atleast one of the sample means is different from another

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Types of ANOVA

One-Way or Two-Way refers to the number of independent variables (IVs) in your Analysis of Variance test.

One Way Analysis: When we are comparing more than three groups based on one factor variable, then it said to be one way analysis of variance (ANOVA). For example, if we want to compare whether or not the mean output of three workers is the same based on the working hours of the three workers. One-way has one independent variable (with 2 levels), for example: brand of cereal

Two Way Analysis: When factor variables are more than two, then it is said to be two way analysis of variance (ANOVA). For example, based on working condition and working hours, we can compare whether or not the mean output of three workers is the same. Two-way has two independent variables (it can have multiple levels), for example: brand of cereal, calories

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Hypothesis of One-Way ANOVA

$$H_0 : \mu_1 = \mu_2 = \mu_3 = \ldots = \mu_c$$

- All population means are equal i.e., no factor effect (no variation in means among groups)

$$H_1 : \text{Not all of the population means are equal}$$

- At least one population mean is different i.e., ther
- At least one population mean is different i.e., there is a factor effect

(Does not mean that all population means are different, some pairs may be the same)

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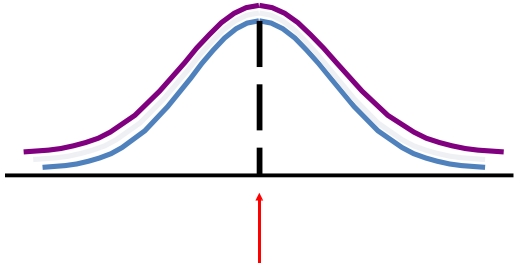
One-Way ANOVA

$$H_0 : \mu_1 = \mu_2 = \mu_3 = \ldots = \mu_c$$

$$H_1 : \text{Not all } \mu_j \text{ are equal}$$

The Null Hypothesis is True All Means are the same:
(No Factor Effect)



$$\mu_1 = \mu_2 = \mu_3$$

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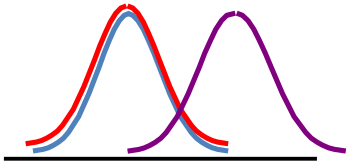
One-Way ANOVA

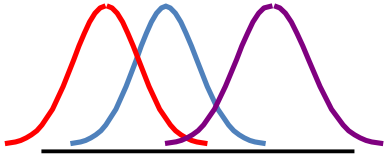
$$H_0 : \mu_1 = \mu_2 = \mu_3 = \ldots = \mu_c$$

$$H_1 : \text{Not all } \mu_j \text{ are equal}$$

The Null Hypothesis is NOT true
At least one of the means is different
(Factor Effect is present)



$\mu_1 = \mu_2 \neq \mu_3$



$\mu_1 \neq \mu_2 \neq \mu_3$

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Partitioning the Variation

Total variation can be split into two parts:

$$SST = SSB + SSW$$

SST = Total Sum of Squares (Total variation)
SSB = Sum of Squares Between Groups (Among-group variation)
SSW = Sum of Squares Within Groups (Within-group variation)

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Partitioning the Variation

$$SST = SSB + SSW$$

Total Variation = the aggregate variation of the individual data values across the various factor levels (SST)

Between-Group Variation = variation between the factor sample means (SSB)

Within-Group Variation = variation that exists among the data values within a particular factor level (SSW)

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Partition of Total Variation

Total Variation (SST)

= **Variation Due to Factor (SSB)** + **Variation Due to Random Error (SSW)**

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Total Sum of Squares

$$SST = SSB + SSW$$

$$SST = \sum_{j=1}^{c} \sum_{i=1}^{n_j} (X_{ij} - \bar{\bar{X}})^2$$

Where:

SST = Total sum of squares

c = number of groups or levels

n_j = number of observations in group j

X_{ij} = i^{th} observation from group j

$\bar{\bar{X}}$ = grand mean (mean of all data values)

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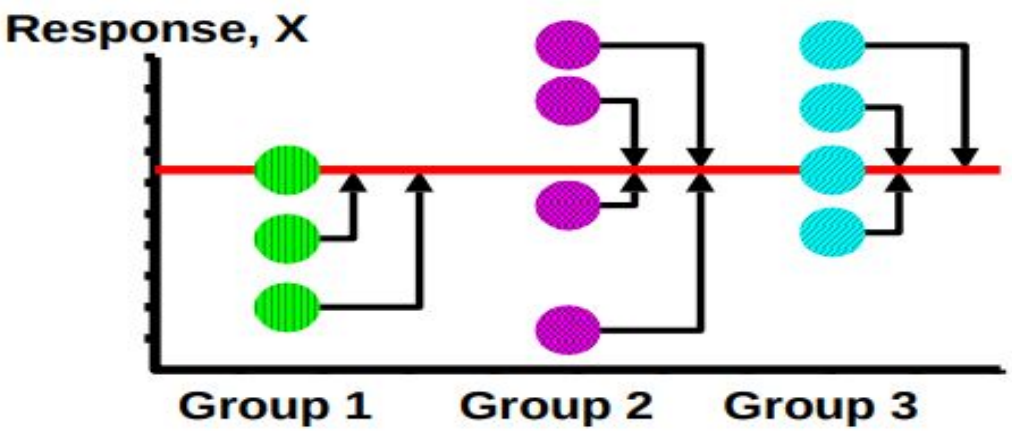
Total Variation

$$SST = (X_{11} - \overline{\overline{X}})^2 + (X_{12} - \overline{\overline{X}})^2 + \cdots + (X_{cn_c} - \overline{\overline{X}})^2$$



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Between-Group Variation

$$SST = SSB + SSW$$

$$SSB = \sum_{j=1}^{c} n_j (\overline{X}_j - \overline{\overline{X}})^2$$

Where:

SSB = Sum of squares between groups

c = number of groups

n_j = sample size from group j

X_j = sample mean from group j

X = grand mean (mean of all data values)

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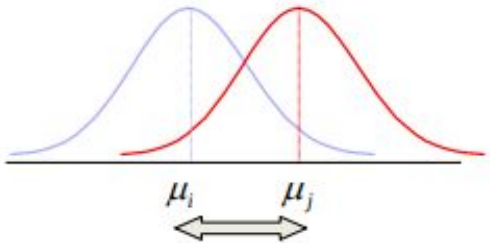
Between-Group Variation

$$SSB = \sum_{j=1}^{c} n_j (\overline{X}_j - \overline{\overline{X}})^2$$

Variation Due to Differences Between Groups



$$MSB = \frac{SSB}{c-1}$$

Mean Square Between = SSB/degrees of freedom

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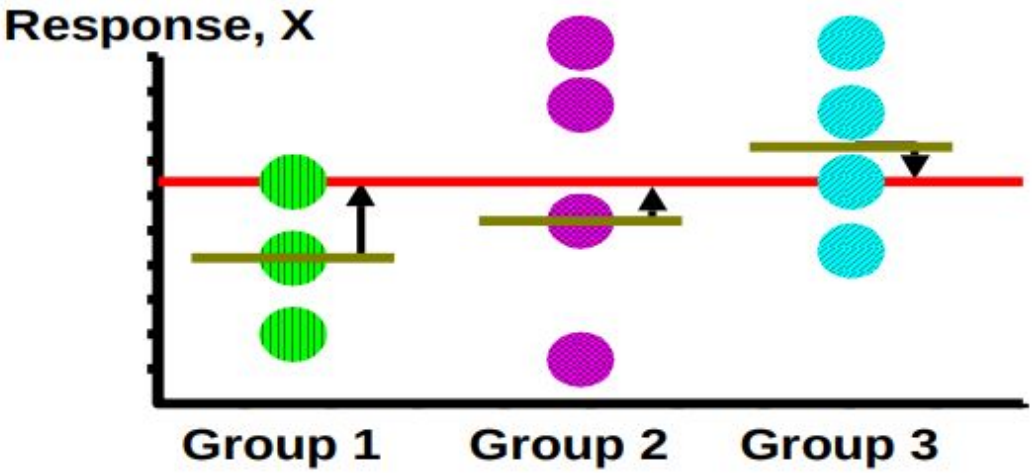
Between-Group Variation

$$SSB = n_1(\overline{X}_1 - \overline{\overline{X}})^2 + n_2(\overline{X}_2 - \overline{\overline{X}})^2 + \cdots + n_c(\overline{X}_c - \overline{\overline{X}})^2$$



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Within-Group Variation

$$SST = SSB + SSW$$

$$SSW = \sum_{j=1}^{c} \sum_{i=1}^{n_j} (X_{ij} - \bar{X}_j)^2$$

Where:

SSW = Sum of squares within groups

c = number of groups

n_j = sample size from group j

X_j = sample mean from group j

X_{ij} = i^{th} observation in group j

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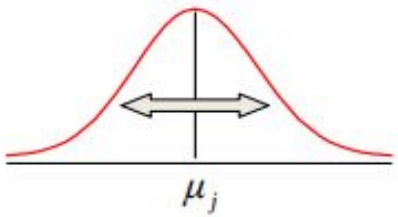
Within-Group Variation

$$SSW = \sum_{j=1}^{c} \sum_{i=1}^{n_j} (X_{ij} - \overline{X}_j)^2$$

Summing the variation within each group and then adding over all groups



$$MSW = \frac{SSW}{n - c}$$

Mean Square Within = SSW/degrees of freedom

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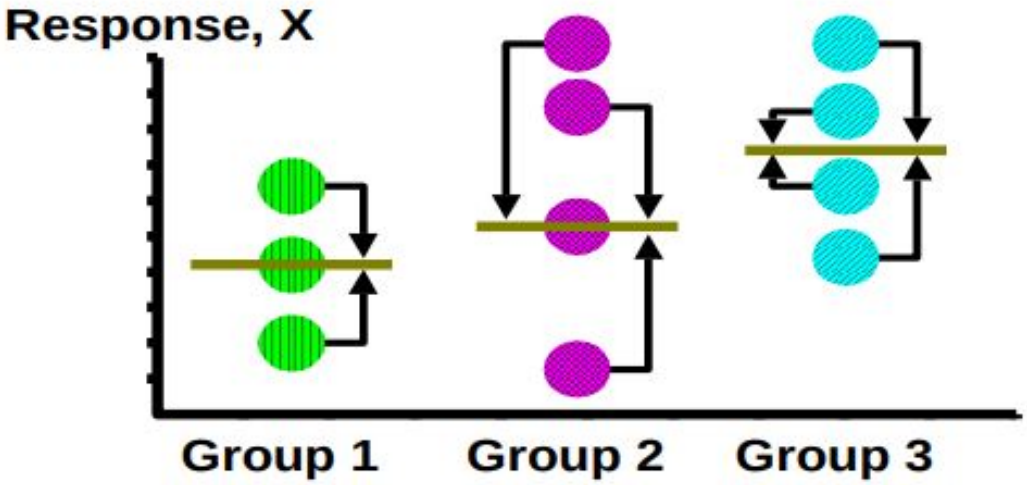
Within-Group Variation

$$SSW = (X_{11} - \overline{X}_1)^2 + (X_{12} - \overline{X}_2)^2 + \cdots + (X_{cn_c} - \overline{X}_c)^2$$



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Obtaining the Mean Squares

The Mean Squares are obtained by dividing the various sum of squares by their associated degrees of freedom

$$MSB = \frac{SSB}{c-1}$$

Mean Square Between
(d.f. = c-1)

$$MSW = \frac{SSW}{n-c}$$

Mean Square Within
(d.f. = n-c)

$$MST = \frac{SST}{n-1}$$

Mean Square Total

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One-Way ANOVA Table

SOURCE OF VARIATION	DEGREES OF FREEDOM	SUM OF SQUARES	MEAN SQUARE (VARIANCE)	F
Between Groups	c - 1	SSB	$MSB = \frac{SSA}{c - 1}$	$F_{STAT} = \frac{MSB}{MSW}$
Within Groups	n - c	SSW	$MSW = \frac{SSW}{n - c}$	
Total	n - 1	SST		

c = number of groups

n = sum of the sample sizes from all groups

df = degrees of freedom

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One-Way ANOVA - F Test Statistic

H_0: $\mu_1 = \mu_2 = \cdots = \mu_c$

H_1: At least two population means are different

Test statistic

$$F_{STAT} = \frac{MSB}{MSW}$$

MSB is mean squares between groups
MSW is mean squares within groups

Degrees of freedom

$df_1 = c - 1$ (c = number of groups)
$df_2 = n - c$ (n = sum of sample sizes from all populations)

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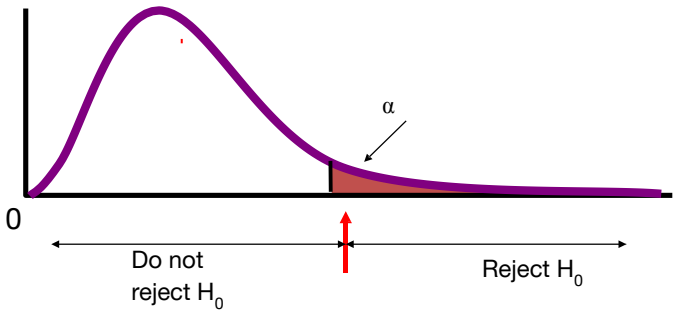
Interpreting One-Way ANOVA - F Statistic

The F statistic is the ratio of the between estimate of variance and the within estimate of variance

- The ratio must always be positive
- $df_1 = c - 1$ will typically be small
- $df_2 = n - c$ will typically be large

Decision Rule:

- Reject H_0 if $F_{STAT} > F_\alpha$, otherwise do not reject H_0



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One-Way ANOVA - F Test Example

You want to see if three different golf clubs yield different distances. You randomly select five measurements from trials on an automated driving machine for each club. At the 0.05 significance level, is there a difference in mean distance?

CLUB 1	CLUB 2	CLUB 3
254	234	200
263	218	222
241	235	197
237	227	206
251	216	204

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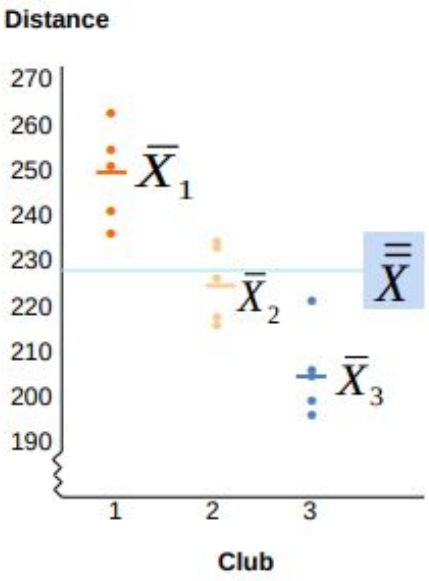
One-Way ANOVA Example: Scatter Plot

CLUB 1	CLUB 2	CLUB 3
254	234	200
263	218	222
241	235	197
237	227	206
251	216	204

$$\bar{x}_1 = 249.2 \mid \bar{x}_2 = 226.0 \mid \bar{x}_3 = 205.8$$

$$\bar{\bar{x}} = 227.0$$



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One-Way ANOVA Example Computations

CLUB 1	CLUB 2	CLUB 3
254	234	200
263	218	222
241	235	197
237	227	206
251	216	204

$X_1 = 249.2 \quad n_1 = 5$

$X_2 = 226.0 \quad n_2 = 5$

$X_3 = 205.8 \quad n_3 = 5$

$n = 15$

$\bar{\bar{X}} = 227.0 \quad c = 3$

$$SSB = 5\,(249.2 - 227)^2 + 5\,(226 - 227)^2 + 5\,(205.8 - 227)^2 = 4716.4$$

$$SSW = (254 - 249.2)^2 + (263 - 249.2)^2 + \cdots + (204 - 205.8)^2 = 1119.6$$

MSB = 4716.4 / (3-1) = 2358.2

MSW = 1119.6 / (15-3) = 93.3

$$F_{STAT} = \frac{2358.2}{} = 25.275$$

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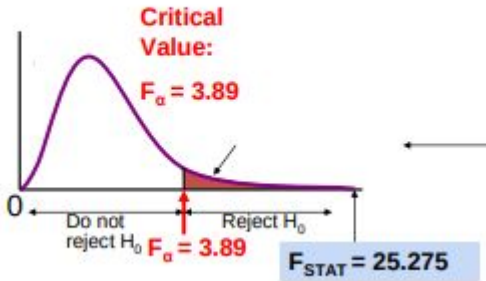
One-Way ANOVA Example Computations

H_0: $\mu_1 = \mu_2 = \mu_3$
H_1: μ_j not all equal
$\alpha = 0.05$
$df_1 = 2$ $df_2 = 12$

Test Statistic:

$$F_{STAT} = \frac{MSB}{MSW} = \frac{2358.2}{93.3} = 25.275$$



Decision:

Reject H_0 at $\alpha = 0.05$

Conclusion:

There is evidence that at least one μ_j differs from the rest

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One-Way ANOVA Example Computations

SUMMARY						
Groups	*Count*	*Sum*	*Average*	*Variance*		
Club 1	5	1246	249.2	108.2		
Club 2	5	1130	226	77.5		
Club 3	5	1029	205.8	94.2		
ANOVA						
Source of Variation	*SS*	*df*	*MS*	*F*	*P-value*	*F crit*
Between Groups	**4716.4**	**2**	**2358.2**	**25.275**	0.0000	**3.89**
Within Groups	**1119.6**	**12**	**93.3**			
Total	5836.0	14				

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ANOVA Assumptions Levene's Test

- Tests the assumption that the variances of each population are equal.
- First, define the null and alternative hypotheses:
  - $H_0: \sigma^2_1 = \sigma^2_2 = \cdots = \sigma^2_c$
  - H_1: Not all σ^2_j are equal
- Second, compute the absolute value of the difference between each value and the median of each group.
- Third, perform a one-way ANOVA on these absolute differences.

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What to do with a Significant ANOVA result

If the ANOVA is significant and the null hypothesis is rejected, the only valid inference that can be made is that at least one mean is different from at least one other mean

- The ANOVA does not reveal which population means differ from which others

- Some questions we might want to ask:
  - Is one Group much better than all the other Group?
  - Is one Group much worse than all the other Group?

- Only think about investigating differences between individual groups when the overall comparison of groups (ANOVA) is significant, or that you had intended particular comparisons at the outset

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Question

Ryan is a production manager at an industry manufacturing alloy wires. They have 4 machines - A, B, C and D.

Ryan wants to study whether all the machines have equal efficiency based on the tensile strength of the alloy wire.

Is it possible to test his claim?

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Solution

The trivial solution is conducting multiple t tests. However performing multiple t tests has an effect on the type I error.

As the number of t-tests increases the probability of at least one type I error increases.

However, it is possible to test Ryans claim by using one way analysis of variance (one way ANOVA) where the probability of type I error does not change

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Multiple t tests and type I error

- For a true null hypothesis, the probability of not obtaining a significant result is 0.95 if the $\alpha = 0.05$

- Say you conduct the t-test twice, the probability of not obtaining one or more significant result is 0.95 x 0.95 = 0.9025

- Thus the probability of at least one type error is 1-0.9025 = 0.0975 (for two t-tests)

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Multiple t tests and type I error

Number of t tests	Probability of not obtaining one or more significant result	Probability of at least one type I error
3 t tests	0.95 x 0.95 x 0.95 = 0.857	0.143
4 t tests	0.95 x 0.95 x 0.95 x 0.95 = 0.815	0.185
5 t tests	0.95 x 0.95 x 0.95 x 0.95 x 0.95 = 0.774	0.226

As the number of tests increase the probability of at least one type error also increases

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One way ANOVA

- A t-test is used when two unpaired data are compared
- To test the equality of population means for two or more unrelated samples ANOVA technique is used
- Each group is considered to be a treatment
- It is based on the F-distribution

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One way ANOVA - assumption

- The samples should be independent
- Each sample should be from normally distributed population
- The population variance of the samples should be equal (homoscedastic)

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One way ANOVA

- The null hypothesis to be tested is

H_0: The averages of all treatments are same.
i.e. $\mu_1 = \mu_2 = \ldots = \mu_n$

Against

H_1: At the least one treatment has a different average

- Failing to reject H_0, implies that all treatments have the same average

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One way ANOVA

- Suppose Ryan collects data for tensile strength of wires produced by each machine
- It is said there are 4 treatments (t = 4)
- Each treatment has 5 observations (n_i = 5) where i = 1, 2, …, t
- Total number of observations is given by N

$$N = \sum_i^t n_i$$

A	B	C	D
68.7	62.7	55.9	80.7
75.4	68.5	56.1	70.3
70.9	63.1	57.3	80.9
79.1	62.2	59.2	85.4
78.2	60.3	50.1	82.3

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One way ANOVA

- Let μ_i (i=1, 2, …, t) denote the average strength due to each machine

- For our example, t = 4

- The test hypothesis can be written as

H_0: $\mu_1 = \mu_2 = \mu_3 = \mu_4$ Against H_1: At least μ_i is different

A	B	C	D
68.7	62.7	55.9	80.7
75.4	68.5	56.1	70.3
70.9	63.1	57.3	80.9
79.1	62.2	59.2	85.4
78.2	60.3	50.1	82.3

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One way ANOVA

- In one way ANOVA, the entire population variance is split into two component
  - Variation within treatment
  - Variation between treatment
- Total variation = Within treatment variation + Between treatment variation

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Total variation

- It is the total sum of squares (TSS)
- Let x_{ij} be the observations in the ith treatment and j^{th} row
- $\bar{x}_{..}$ is the grand mean, i.e. the mean of all observations
- The total variation is given by

$$TSS = \sum_i^t \sum_j^{n_i} (x_{ij} - \bar{x}_{..})^2$$

Summation over all treatments

Summation over all observation in the treatment

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Within treatment variation

- It is the treatment sum of squares (TrSS)
- Let x_i be the observations in the ith treatment with n_i in observation in each treatment and $\bar{x}_{i.}$ is the mean over i^{th} treatment
- $\bar{x}_{..}$ is the grand mean, i.e. the mean of all observations
- The treatment variation is given by

$$TrSS = \sum_i^t \sum_j^{n_i} n_i(\bar{x}_{i.} - \bar{x}_{..})^2$$

Summation over all treatments

Summation over all observation in the treatment

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Between treatment variation

- It is the error sum of squares (ESS)
- Let $x_{i.}$ be the observations in the ith treatment and $\bar{x}_{i.}$ is the mean over j^{th} row
- $\bar{x}_{..}$ is the grand mean, i.e. the mean of all observations
- The error sum of squares is given by

$$ESS = \sum_i^t \sum_j^{n_i} (x_{ij} - \bar{x}_{i.})^2$$

Summation over all treatments

Summation over all observation in the treatment

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Error sum of squares

During problem solving, the error sum of squares is obtained as:

$$\mathrm{ESS} = \mathrm{TSS} - \mathrm{TrSS}$$

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One way ANOVA

- The test statistic is given by

$$\text{F-ratio} = \frac{\frac{TrSS}{df_{Tr}}}{\frac{ESS}{df_e}} = \frac{\text{MTrSS}}{\text{MESS}}$$

Mean Treatment Sum of Squares

Mean Error Sum of Squares

- Under H_0, the test statistic follows F-distribution with (df_{Tr}, df_e) degrees of freedom

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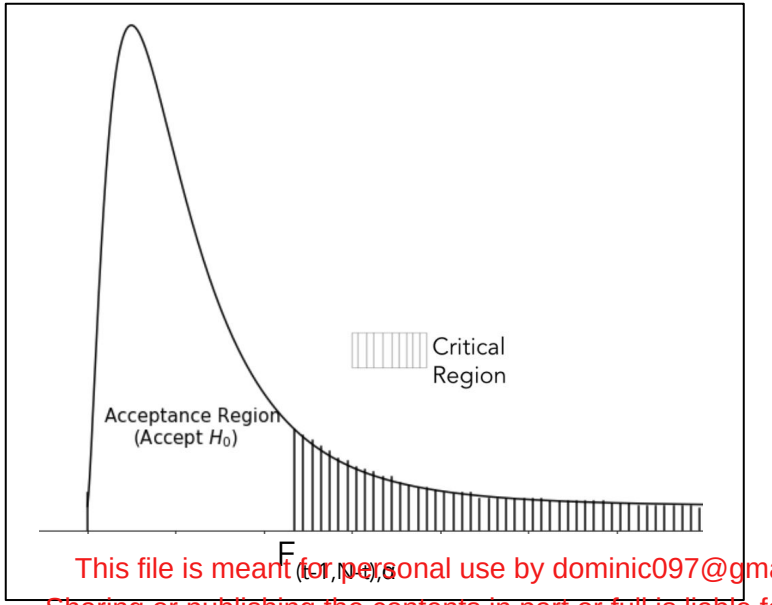
One way ANOVA

Decision Rule: If $F_{cal} \geq F_{(t-1,N-t),\alpha}$ or p-value $\leq \alpha$, then we reject H_0 at α% level of significance



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One way ANOVA

To ease the entire computational process, an ANOVA table is prepared as follows:

Source of variation	Degrees of freedom	Sum of Squares	Mean Sum of Squares	F-ratio
Treatment	t-1	TrSS	s^2_t	$\frac{s_t^2}{s_e^2}$
Error	N-t	ESS	s^2_e	
Total	N-1	TSS		

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One way ANOVA - procedure

1. State the hypothesis to be tested
2. Compute the sum of squares
   a. The total sum of squares, TSS = $\sum_{j=1}^{t}\sum_{i=1}^{ni} (x_{ij} - \bar{x}_{..})^2$
   b. The treatment sum of squares TrSS = $\sum_{j=1}^{t}\sum_{i=1}^{ni} n_i(x_{ij} - \bar{x}_{i.})^2$
   c. The Error sum of squares, ESS = TSS - TrSS
3. Compute mean sum of squares
   a. s^2_t = Mean group sum of squares (MTrSS) = TrSS/(t-1)
   b. s^2_e = Mean error sum of squares (MESS) = ESS/(N-t)

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One way ANOVA - procedure

4. Compute the F-ratio

$$\text{F-ratio} = \frac{\text{MTrSS}}{\text{MESS}} = \frac{s_t^2}{s_e^2}$$

4. Prepare the ANOVA table

5. Write the decision and conclusion accordingly

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One way ANOVA

Question:

Ryan is a production manager at an industry manufacturing alloy seals. They have 4 machines - A, B, C and D. Ryan wants to study whether all the machines have equal efficiency.

Ryan collects data of tensile strength (in N/m^2) from all the 4 machines as given.

Test at 5% level of significance.

A	B	C	D
68.7	62.7	55.9	80.7
75.4	68.5	56.1	70.3
70.9	63.1	57.3	80.9
79.1	62.2	59.2	85.4
78.2	60.3	50.1	82.3

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One way ANOVA

Solution:

Ryan is a production manager at an industry manufacturing alloy seals. They have 4 machines - A, B, C and D

Let μ_1 be the average tensile strength due to machine A
μ_2 be the average tensile strength due to machine B
μ_3 be the average tensile strength due to machine C
μ_4 be the average tensile strength due to machine D

To test, H_0: $\mu_1 = \mu_2 = \mu_3 = \mu_4$ Against H_1: At least one μ_i is different (i=1, 2, 3, 4)

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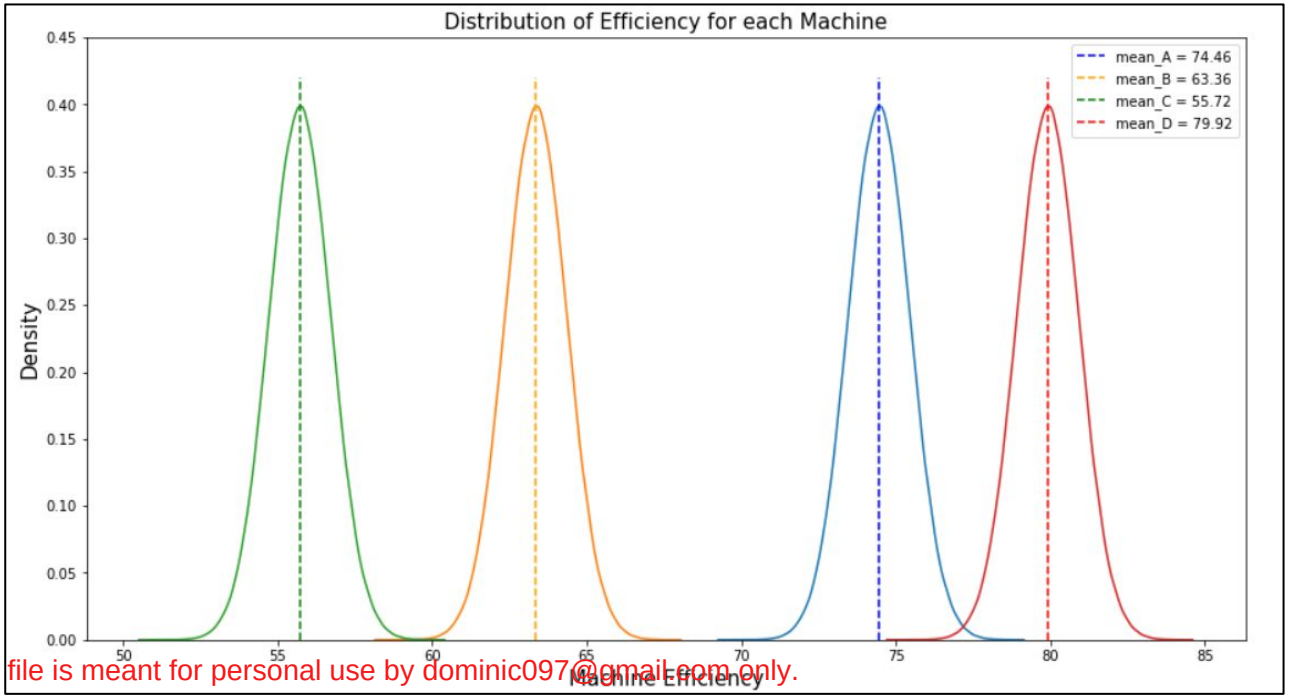
One way ANOVA

The plot shows the difference between the average efficiency for each machine, which indicates the rejection of H_0.



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One way ANOVA

Solution:

The grand mean:

$$\bar{x}_{..} = \frac{68.7+62.7+...+50.1+82.3}{20} = 68.365 \ \text{N/m}^2$$

The total sum of squares:

$$TSS = \sum_i^t \sum_j^{n_i} (x_{ij} - \bar{x}_{..})^2$$

$$= (68.7 - 69.115)^2 + \ldots + (81.12 - 69.115)^2$$

$$= 2074.1255 \ (\text{N/m}^2)^2$$

A	B	C	D
68.7	62.7	55.9	80.7
75.4	68.5	56.1	70.3
70.9	63.1	57.3	80.9
79.1	62.2	59.2	85.4
78.2	60.3	50.1	82.3

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One way ANOVA

Solution:

The treatment sum of squares is

$$TrSS = \sum_i^t \sum_j^{n_i} n_i(\overline{x}_{i.} - \overline{x}_{..})^2$$

$$= 5(74.46 - 68.365)^2 + \ldots + 5(79.92 - 68.365)^2$$

$$= 1778.0655 \ (\mathrm{N/m}^2)^2$$

	A	B	C	D
	68.7	62.7	55.9	80.7
	75.4	68.5	56.1	70.3
	70.9	63.1	57.3	80.9
	79.1	62.2	59.2	85.4
	78.2	60.3	50.1	82.3
$\sum x_i$	372.3	316.8	278.6	399.6
$\bar{x}_{i.}$	74.46	63.36	55.72	79.92

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One way ANOVA

Solution:

The error sum of squares can also be calculated as

$$ESS = \sum_i^t \sum_j^{n_i} (x_{ij} - \bar{x}_{i.})^2$$

$$= (68.7 - 74.46)^2 + \ldots + (82.3 - 79.92)^2$$

$$= 296.06$$

	A	B	C	D
	68.7	62.7	55.9	80.7
	75.4	64.5	56.1	80.3
	70.9	63.1	57.3	80.9
	79.1	59.2	55.2	81.4
	78.2	60.3	50.1	82.3
$\bar{x}_{i.}$	74.46	63.36	55.72	79.92

Or can be obtained as $ESS = TSS - TSS = 296.06$

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One way ANOVA

Solution:

Source of variation	Degrees of freedom	Sum of Squares	Mean Sum of Squares	F-ratio
Treatment	t-1 = 4-1 =3	TrSS = 1778.0655	$s_t^2 = \frac{21778.0655}{3} = 592.6885$	$\frac{s_t^2}{s_e^2} = 32.031$
Error	N-t = 20-4 =16	ESS = 296.06	$s_e^2 = \frac{269.06}{16} = 18.50375$	
Total	N-1 = 20 -1 =19	TSS = 2241.5255		

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One way ANOVA

Solution:

From the F-table we have $F_{(3,16),0.05} = 3.24$

Since $3.24 < 32.03$, we reject H_0.

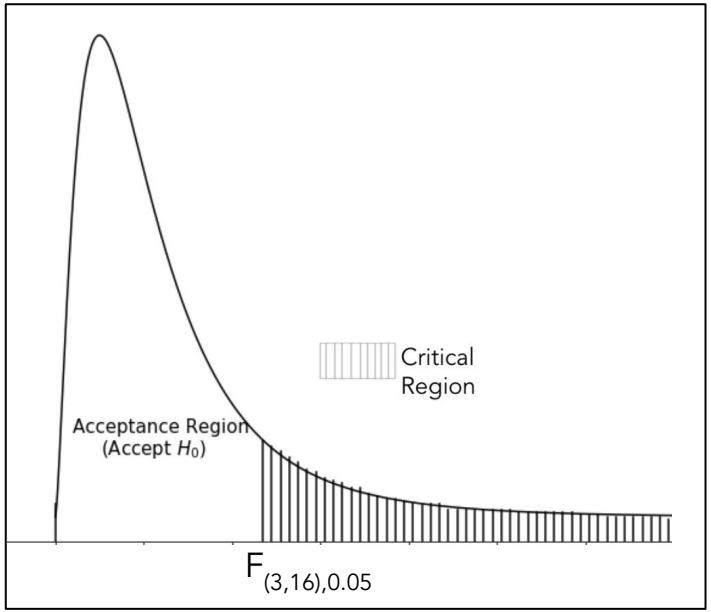

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Table

Question:

A researcher tests **three fertilizers** (A, B, C) to see whether they produce different mean crop yields (kg). The yields (per plot) are:

- **Fertilizer A (n_1 = 4):** 20, 22, 19, 24
- **Fertilizer B (n_2 = 3):** 30, 28, 33
- **Fertilizer C (n_3 = 5):** 25, 27, 26, 28, 24

Test at **α = 0.05** whether the mean yields differ.

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Step 1 — Group sums, means and totals

Group A:

- $n_1 = 4, \sum X_A = 20 + 22 + 19 + 24 = 85$
- $\bar{X}_A = 85/4 = 21.25$

Group B:

- $n_2 = 3, \sum X_B = 30 + 28 + 33 = 91$
- $\bar{X}_B = 91/3 \approx 30.3333$

Group C:

- $n_3 = 5, \sum X_C = 25 + 27 + 26 + 28 + 24 = 130$
- $\bar{X}_C = 130/5 = 26.0$

Total:

- $N = 4 + 3 + 5 = 12$
- Grand sum $\sum X = 85 + 91 + 130 = 306$
- Grand mean $\bar{X}_{GM} = 306/12 = 25.5$

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Step 2 — Between-group sum of squares (SSB)

Formula:

$$SSB = \sum_i n_i(\bar{X}_i - \bar{X}_{GM})^2$$

Compute each term:

- A: $4(21.25 - 25.5)^2 = 4(-4.25)^2 = 4(18.0625) = 72.25$
- B: $3(30.3333 - 25.5)^2 = 3(4.8333)^2 \approx 3(23.361111) \approx 70.083333$
- C: $5(26.0 - 25.5)^2 = 5(0.5)^2 = 5(0.25) = 1.25$

$$\boxed{SSB \approx 72.25 + 70.083333 + 1.25 = 143.583333}$$

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Step 3 — Within-group sum of squares (SSW)

Compute sum of squared deviations within each group:

Group A (mean = 21.25):

$$(20-21.25)^2+(22-21.25)^2+(19-21.25)^2+(24-21.25)^2=1.5625+0.5625+5.0625+7.5625=14.75$$

Group B (mean ≈ 30.3333):

$$(30-30.3333)^2+(28-30.3333)^2+(33-30.3333)^2\approx 0.1111+5.4444+7.1111=12.6666$$

Group C (mean = 26.0):

$$(25-26)^2+(27-26)^2+(26-26)^2+(28-26)^2+(24-26)^2=1+1+0+4+4=10$$

$$\boxed{SSW=14.75+12.666666+10=37.416666\ (\approx 37.42)}$$

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Step 4 — Total sum of squares (TSS)

$$TSS = SSB + SSW = 143.583333 + 37.416666 \approx \boxed{181.0}$$

Step 5 — Degrees of freedom, Mean Squares, F

- $k = 3$ groups.
- $df_{between} = k - 1 = 2$
- $df_{within} = N - k = 12 - 3 = 9$

Mean squares:

$$MSB = SSB/df_{between} = 143.583333/2 = 71.7916665$$

$$MSW = SSW/df_{within} = 37.416666/9 \approx 4.1574073$$

F statistic:

$$F = \frac{MSB}{MSW} \approx \frac{71.7916665}{4.1574073} \approx \boxed{17.27}$$

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Source	SS	df	MS	F
Between (A,B,C)	143.583	2	71.792	17.27
Within (Error)	37.417	9	4.157	
Total	181	11		

Step 6 — Decision (compare with critical F)

- At α = 0.05, $df_1 = 2$, $df_2 = 9$, the critical value $F_{0.05}(2, 9) \approx 4.26$.
- Since $F_{calc} \approx 17.27 > 4.26$, **reject** H_0.

You can also report the p-value: with F ≈ 17.27 and df (2,9), **p ≪ 0.01** (very small), so strong evidence to reject H_0.

Conclusion:

There is **strong evidence** that mean yields differ between the three fertilizers (p < 0.01). In other words, **not all fertilizer means are equal** — at least one group's mean yield is significantly different from the others.

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Thanks!...

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